

Mendel's Laws

Note: In OneNote, set this worksheet as background. Use "Export data" in the simulation and insert the result in the marked space.

Aim of the work

Use the simulation to investigate monohybrid and dihybrid crosses. Compare Punnett-square expectations with simulated results and formulate the three Mendelian laws.

How to work with the simulation

- 1) Choose the trait(s): seed colour (Y/y), seed shape (R/r) or both.**
- 2) Choose the cross and set the parent genotypes.**
- 3) Predict the gametes first; use support mode only if needed.**
- 4) Generate the Punnett square and note expected genotype and phenotype ratios.**
- 5) Choose n, simulate the cross and compare expectation with observation.**
- 6) Insert important exports or screenshots into this worksheet.**

Memory aid: Punnett square and gametes

A Punnett square combines the parents' gametes. Each cell shows a possible offspring genotype. Homozygous parents form one allele variant per gene, heterozygous parents two; dihybrid crosses combine several variants.

Key terms and preparation

Term	Meaning / example
genotype	Allele combination of a trait, e.g. Yy
phenotype	Visible expression of a trait, e.g. yellow
dominant	Prevails in the heterozygous state
recessive	Is visible only without a dominant allele
homozygous	Two identical alleles, e.g. YY or yy
heterozygous	Two different alleles, e.g. Yy
Gameten / gametes	Contain only one allele per gene
Punnett-Quadrat/ Crosssquadrat/ Crosssschema	Scheme for determining expected offspring

Short exercise on gamete formation

Parent genotype	Possible gametes
YY	
Yy	
RR	
YyRr	

Experiment A - 1st law (uniformity)

Assignment: Cross two homozygous parents with one differing trait. Examine F1.

Planning

Trait	Parent 1 (genotype)	Parent 2 (genotype)	F1 genotype	F1 phenotype

Procedure: Simulate F1 with n = 50 and n = 200.

n	Observed phenotypeen	Observed genotypeen	Vergleich mit der expectation / note
50			
200			

Export: Punnett square or F1 results

Insert export here

Formulate the 1st law:

Experiment B - 2nd law (segregation)

Assignment: Cross two F1 individuals. Compare expected and simulated F2 ratios.

Planning

Cross (F1 x F1)	Exp. phenotype	Exp. genotype

Procedure: Simulate F2 with n = 50, n = 200 and n = 1000.

n	Observed phenotypeen	Observed genotypeen	Vergleich mit der expectation / note
50			
200			
1000			

Export: F2 results

Insert export here

Formulate the 2nd law:

Experiment C - 3rd law (independent assortment)

Assignment: Investigate a dihybrid cross (colour + shape). Use homozygous parents, then F1 x F1.

Preparation: create F1 with two traits

Parent 1 (genotype)	Parent 2 (genotype)	Trait 1	Trait 2	Expectation for the F1

F2: create a Punnett square and note the expected ratio

Cross (F1 x F1)	Expected phenotype- ratio	Notes / special features

Procedure: Simulate F2 with n = 200 and n = 1000.

n	Observed phenotypeen	ratio (phenotype)	Deviation
200			
1000			

Export from the simulation: dihybrid cross / result view

Insert export here

Formulate the 3rd law:

Overview of the three laws

Law	Cross of the parents	ratio (phenotype)	ratio (genotype)
1st law			
2nd law			
3rd law of independent assortment			

What role do chance and sample size play?

Use your results to explain independent assortment.
